

Unit II: Linear Algebra for AI and Quantum Systems

Scalars, vectors, matrices, matrix operations, systems of equations, and intro to vector spaces

Learning Goals

- Understand scalars, vectors, and matrices as basic objects for representing data and quantum states.
- Perform common matrix operations used in machine learning, data science, and quantum computing.
- Solve and interpret systems of linear equations.
- Develop an introductory understanding of vector spaces, subspaces, span, basis, and dimension.
- Connect linear algebra ideas to feature vectors, transformations, neural networks, and qubits.

1. Why Linear Algebra Matters

Linear algebra is the language of structured numerical information. In artificial intelligence, an image, word, user, document, or sensor reading is often represented as a vector of numbers. In quantum systems, the state of a qubit or a larger quantum register is also represented using vectors, usually with complex-valued entries called amplitudes.

Matrices describe transformations. They can rotate vectors, project data into lower dimensions, mix features in a neural network layer, or evolve a quantum state through a gate. Because both AI and quantum theory depend on transformations of vectors, linear algebra becomes a shared mathematical foundation.

2. Scalars

A scalar is a single number. It has magnitude but no direction. Scalars may be real numbers such as 3, -2.5, and 0.01, or complex numbers such as $2 + 3i$. Scalars are used to scale vectors and matrices, meaning they stretch, shrink, or reverse them.

Scalar multiplication: if c is a scalar and $v = [v_1, v_2, \dots, v_n]$, then $c v = [c v_1, c v_2, \dots, c v_n]$

- In AI, a learning rate such as 0.01 is a scalar that controls the size of an update step.
- In quantum computing, probability amplitudes may be complex scalars attached to basis states.
- Multiplying by 0 collapses a vector to the zero vector; multiplying by -1 reverses its direction.

3. Vectors

A vector is an ordered list of numbers. Geometrically, a vector can represent a directed arrow in space. Computationally, it can represent a record of features. For example, a student may be represented by marks in several subjects, an image by pixel intensities, and a word by an embedding vector learned from text.

$$v = [v_1, v_2, \dots, v_n]$$

$$\text{length/dimension} = n$$

Vector Operations

- Addition: vectors of the same dimension are added component by component.
- Subtraction: subtract corresponding components to compare displacement or difference.
- Scalar multiplication: multiply every component by the same scalar.
- Magnitude or norm: measures the length of a vector.
- Dot product: measures alignment or similarity between two vectors.

$$u + v = [u_1 + v_1, \dots, u_n + v_n]$$

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

$$||v|| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

$$\cos(\theta) = (u \cdot v) / (||u|| ||v||)$$

The dot product is especially important in AI. If two feature vectors point in similar directions, their dot product is large. Search engines, recommender systems, and embedding models often use dot products or cosine similarity to compare meaning or preference.

Feature Vectors

A feature vector is a numerical representation of an object. A house may be represented as [area, bedrooms, age, distance_to_city]. A patient record may be represented as [heart_rate, blood_pressure, glucose_level]. Once data is converted into vectors, algorithms can compare, classify, cluster, and predict using mathematical operations.

- Good features preserve useful information and remove unnecessary noise.
- Feature scaling is often needed because one large-valued feature can dominate distance or dot product calculations.
- High-dimensional vectors can represent complex objects such as images, speech, and natural language.

4. Matrices

A matrix is a rectangular array of numbers arranged in rows and columns. A matrix with m rows and n columns is called an m by n matrix. Matrices can store datasets, coefficients of equations, images, graph connections, and linear transformations.

$$A = [a_{ij}]$$

$$A \text{ has } m \text{ rows and } n \text{ columns}$$

$$\text{shape}(A) = m \times n$$

Matrix type	Meaning
Row matrix	A matrix with one row.
Column matrix	A matrix with one column; often treated as a vector.
Square matrix	Same number of rows and columns.
Zero matrix	All entries are zero.
Identity matrix	Square matrix with 1 on the diagonal and 0 elsewhere.

The identity matrix acts like the number 1 for matrix multiplication. If dimensions match, multiplying by the identity leaves a vector or matrix unchanged.

$$I v = v \quad A I = A \quad I A = A$$

5. Matrix Operations

Addition and Scalar Multiplication

Two matrices can be added only when they have the same shape. Addition is component-wise. Scalar multiplication multiplies every entry by the scalar.

$$(A + B)_{ij} = A_{ij} + B_{ij} \quad (cA)_{ij} = c A_{ij}$$

Matrix-Vector Multiplication

Matrix-vector multiplication transforms a vector into another vector. Each output component is a dot product between one row of the matrix and the input vector.

If A is $m \times n$ and x is $n \times 1$, then Ax is $m \times 1$

- In AI, a linear model computes $y = w \cdot x + b$, where x is the feature vector and w is the weight vector.
- A neural network layer often computes $y = Wx + b$ before applying a nonlinear activation.
- In quantum computing, a gate is represented by a matrix that acts on a quantum state vector.

Matrix-Matrix Multiplication

Matrix multiplication combines transformations. If A is m by n and B is n by p , then AB is m by p . The entry in row i and column j of AB is the dot product of row i of A with column j of B .

$$(AB)_{ij} = \text{sum over } k \text{ of } A_{ik} B_{kj}$$

- Matrix multiplication is generally not commutative: AB is usually not equal to BA .
- The order matters because applying transformation A then B is not always the same as applying B then A .
- Deep learning uses repeated matrix multiplications to move information through layers.

Transpose

The transpose of a matrix swaps rows and columns. If A is m by n , then A^T is n by m . Transpose operations are common in least squares, gradients, covariance matrices, and inner products.

$$(A^T)_{ij} = A_{ji} \quad (AB)^T = B^T A^T$$

6. Systems of Linear Equations

A system of linear equations is a collection of equations where each variable appears only to the first power. Such systems can be written compactly using matrices.

$$a_{11} x_1 + a_{12} x_2 = b_1 \quad a_{21} x_1 + a_{22} x_2 = b_2 \quad \text{Matrix form: } A x = b$$

Solving $Ax = b$ means finding the vector x that satisfies all equations at once. There may be exactly one solution, no solution, or infinitely many solutions.

Case	Interpretation
Unique solution	Equations meet at one point; matrix has enough independent information.
No solution	Equations are inconsistent; for example, parallel lines with different intercepts.
Infinitely many	Equations are dependent; one equation repeats information from another.

Gaussian Elimination

Gaussian elimination solves a system by using row operations to simplify the augmented matrix. The aim is to form an upper triangular or row-echelon matrix, then solve variables by back substitution.

- Swap two rows when needed.
- Multiply a row by a nonzero scalar.
- Add a multiple of one row to another row.

These operations preserve the solution set. In AI and data science, linear systems appear in regression, optimization, graph algorithms, and numerical simulation.

7. Determinant and Inverse

For a square matrix, the determinant is a scalar that gives information about scaling, orientation, and invertibility. If $\det(A) = 0$, the matrix is singular and has no inverse. If $\det(A)$ is nonzero, A has an inverse.

$$\text{For } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \det(A) = ad - bc$$

The inverse of a matrix reverses the transformation. If A has inverse A^{-1} , then $A^{-1}A = I$ and $AA^{-1} = I$. In theory, $x = A^{-1}b$ solves $Ax = b$. In practice, numerical algorithms often solve systems directly rather than explicitly computing the inverse.

$$Ax = b \quad \text{if } A^{-1} \text{ exists, } x = A^{-1}b$$

8. Intro to Vector Spaces

A vector space is a set of vectors where vector addition and scalar multiplication are defined and behave consistently. The usual two-dimensional plane, three-dimensional space, and n -dimensional real coordinate spaces are examples of vector spaces.

Core Ideas

- Closure: adding vectors or multiplying by scalars keeps the result inside the space.
- Zero vector: every vector space contains a vector that acts like zero under addition.
- Additive inverse: for every vector v , there is a vector $-v$ such that $v + (-v) = 0$.
- Linear combination: an expression such as $c_1 v_1 + c_2 v_2 + \dots + c_k v_k$.
- Span: all vectors that can be produced as linear combinations of a given set.
- Basis: a minimal set of vectors that spans the space without redundancy.
- Dimension: the number of vectors in a basis.

$$\text{span}(v_1, v_2, \dots, v_k) = \text{all } c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

Linear Independence

Vectors are linearly independent if none of them can be built from the others. If one vector is a combination of the rest, it is redundant. A basis must be linearly independent and must span the space.

- In $\mathbb{R}^2$, two non-parallel vectors form a basis.
- In $\mathbb{R}^3$, three vectors not lying in the same plane can form a basis.
- In data analysis, dependent features may repeat information and make learning less stable.

9. AI Connections

- Feature vectors represent raw objects numerically so algorithms can process them.
- Matrices store datasets: rows often represent examples and columns often represent features.
- Linear regression uses vectors and matrices to fit a line, plane, or higher-dimensional hyperplane.
- Neural networks use matrix multiplication to combine inputs with learned weights.
- Dimensionality reduction methods such as PCA use linear algebra to find important directions in data.

- Similarity search compares vectors using dot products, distances, or cosine similarity.

10. Quantum Systems Connections

In quantum computing, a state is represented as a vector in a complex vector space. A single qubit has two basis states, written as $|0\rangle$ and $|1\rangle$. A general qubit state is a linear combination of these basis states.

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad \text{where } |\alpha|^2 + |\beta|^2 = 1$$

The numbers α and β are probability amplitudes. Their squared magnitudes give measurement probabilities. Quantum gates are matrices that transform state vectors while preserving total probability.

Classical/AI idea	Quantum analogy
Vector	Quantum state vector.
Basis vectors	Computational basis states such as $ 0\rangle$ and $ 1\rangle$.
Matrix operation	Quantum gate acting on a state.
Norm	State normalization condition.
Inner product	Overlap between quantum states.

- The Pauli-X gate acts like a NOT operation by swapping $|0\rangle$ and $|1\rangle$.
- The Hadamard gate creates superposition from a basis state.
- Multi-qubit systems use larger vector spaces whose dimension grows exponentially with the number of qubits.

Quick Revision Sheet

- A scalar is a single number; a vector is an ordered list of numbers; a matrix is a rectangular array.
- Vector addition and scalar multiplication happen component by component.
- The dot product measures alignment and appears in similarity, projections, and linear models.
- A matrix can represent a dataset, a transformation, or a system of equations.
- Matrix multiplication combines dot products and the order of multiplication matters.
- A system of equations can be written as $Ax = b$.
- Gaussian elimination uses row operations to simplify and solve linear systems.

- A nonzero determinant means a square matrix is invertible.
- A vector space is closed under vector addition and scalar multiplication.
- A basis is an independent spanning set; dimension is the number of basis vectors.
- AI uses vectors for data representation; quantum computing uses vectors for states and matrices for gates.

Practice Questions

- Represent a house using four numerical features and write it as a feature vector.
- Given $u = [1, 2]$ and $v = [3, 4]$, compute $u + v$, $2u$, and $u \cdot v$.
- Explain why two matrices can only be added when they have the same shape.
- If A is 3×2 and x is 2×1 , what is the shape of Ax ?
- Why is matrix multiplication generally not commutative?
- Write a two-equation linear system in matrix form $Ax = b$.
- What are the possible solution cases for a system of linear equations?
- Find $\det(A)$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$. Is A invertible?
- Explain span, basis, and dimension using $\mathbb{R}^2$ as an example.
- Why must a qubit state $\alpha|0\rangle + \beta|1\rangle$ satisfy $|\alpha|^2 + |\beta|^2 = 1$?